SQL Practice Question Set

Table Setup (copy-paste into MySQL)

```
CREATE DATABASE IF NOT EXISTS sql_practice;
USE sql_practice;

CREATE TABLE customers (
    cid INTEGER PRIMARY KEY AUTO_INCREMENT,
    name VARCHAR(100) NOT NULL,
    email VARCHAR(150) NOT NULL UNIQUE,
    age INTEGER CHECK (age > 0),
    city VARCHAR(100)
);

CREATE TABLE orders (
    order_id INTEGER PRIMARY KEY AUTO_INCREMENT,
    cid INTEGER NOT NULL,
    order_date DATETIME DEFAULT CURRENT_TIMESTAMP,
    amount DECIMAL(10,2) NOT NULL,
    CONSTRAINT orders_fk FOREIGN KEY (cid) REFERENCES customers(cid) ON DELETE CASCADE
);

CREATE TABLE smartphones (
    id INTEGER PRIMARY KEY AUTO_INCREMENT,
    brand VARCHAR(50) NOT NULL,
    model VARCHAR(100) NOT NULL,
    price DECIMAL(10,2) NOT NULL,
    rating DECIMAL(3,1),
    screen_size DECIMAL(4,2),
    battery_capacity INTEGER,
    ram INTEGER,
    internal_memory INTEGER,
    primary_camera_rear INTEGER,
    primary_camera_front INTEGER,
    refresh_rate INTEGER,
    ppi INTEGER,
    has_5g BOOLEAN DEFAULT FALSE,
    has_nfc BOOLEAN DEFAULT FALSE,
    has_fast_charging BOOLEAN DEFAULT FALSE,
    processor_brand VARCHAR(50)
);

CREATE TABLE employees (
    employee_id INTEGER PRIMARY KEY AUTO_INCREMENT,
    name VARCHAR(100) NOT NULL,
    manager_id INTEGER,
    salary DECIMAL(10,2),
    department VARCHAR(50)
);
```

Topic 1: DDL (Constraints, ALTER TABLE)

1. Create a table `students` with columns `student_id` (auto-increment PK), `name` (not null), `age` (must be between 6 and 25 using CHECK).

2. Create a table `users` with a named UNIQUE constraint on `email`.

3. Write a query to add a `password` column to the `customers` table.

4. Add a `surname` column to `customers` positioned right after `name`.

5. Drop the `surname` column from `customers`.

6. Modify the `age` column in `customers` to be `NOT NULL`.

7. Add a CHECK constraint on `customers` ensuring `age > 13`.

8. Create the `orders` table with a FOREIGN KEY to `customers(cid)` that cascades on delete.

9. Write a query to rename the `orders` table to `customer_orders`.

10. Drop the foreign key constraint `orders_fk` from `orders`.

Topic 2: DML (INSERT, UPDATE, DELETE)

11. Insert 3 new rows into `customers` with name, email, age, and city.

12. Insert a new row into `smartphones` for a brand of your choice.

13. Update the `price` of all `Samsung` smartphones by reducing 10%.

14. Delete all customers older than 60 from `customers`.

15. Delete all smartphones priced above 200000.

16. Update the `city` of customer with `cid = 5` to `'Mumbai'`.

Topic 3: Aggregate & Scalar Functions

17. Find the minimum and maximum price among all smartphones.

18. Find the price of the costliest Samsung phone.

19. Find the average rating of all Apple phones.

20. Find the number of OnePlus phones.

21. Find the number of distinct brands available.

22. Find the standard deviation of screen sizes.

23. Find the variance of Xiaomi phone prices.

24. Find the difference from average rating for all Samsung phone ratings (use ABS).

25. Round the `ppi` column to 1 decimal place.

26. Find the floor and ceiling of the `rating` column.

27. Find the average battery capacity and average rear camera resolution for phones priced ≥ 100000.

28. Find the average internal memory for phones with refresh rate ≥ 120Hz and front camera ≥ 20MP.

29. Find the number of smartphones with 5G capability.

Topic 4: Sorting (ORDER BY)

30. Find the top 5 Samsung phones with the biggest screen size.

31. Sort all phones in descending order of total cameras (rear + front).

32. Sort smartphone data by `ppi` in decreasing order.

33. Find the phone with the 2nd largest battery capacity.

34. Find the name and rating of the worst-rated Apple phone.

35. Sort phones alphabetically by brand, then by rating in descending order.

36. Sort phones alphabetically by brand, then by price in ascending order.

Topic 5: Grouping (GROUP BY) & HAVING

37. Group smartphones by brand and get count, average price, max rating, average screen size, and average battery capacity.

38. Group smartphones by NFC availability and get average price and rating.

39. Group smartphones by brand and processor brand, get count of models and average rear camera resolution.

40. Find the top 5 most costly phone brands (by average price).

41. Find which brand makes the smallest screen smartphones.

42. Compare average price of 5G phones vs non-5G phones.

43. Group smartphones by brand and find the brand with the highest number of models that have both NFC and 5G.

44. Find all Samsung 5G-enabled smartphones and the average price for NFC vs non-NFC among them.

45. Find the phone name and price of the costliest phone overall.

46. Find the average rating of smartphone brands that have more than 20 phones listed.

47. Find the top 3 brands with the highest average RAM, that have refresh rate ≥ 90Hz and fast charging, excluding brands with fewer than 10 phones.

48. Find the average price of all phone brands with average rating > 70 and more than 10 phones, among all 5G-enabled phones.

Topic 6: Joins

49. Write a query to find all orders along with the customer name who placed them (INNER JOIN).

50. Find all customers and their orders, including customers who haven't placed any order (LEFT JOIN).

51. Find all customers who have never placed an order.

52. Write a query using RIGHT JOIN to list all orders with customer details.

53. Simulate a FULL JOIN between `customers` and `orders` using UNION.

54. Using the `employees` table, write a SELF JOIN to find each employee along with their manager's name.

55. Write a CROSS JOIN example between two small tables of your choice and explain what it returns.

56. Write a join condition using multiple columns (composite key) between two hypothetical tables `order_items` and `products`.

57. Write a 3-table join: `orders` → `customers` → and a hypothetical `order_items` table.

Topic 7: Set Operations

58. Write a UNION query combining customer names from two different city-based queries.

59. Explain the difference between UNION and UNION ALL with an example query for each.

60. Write a query to simulate INTERSECT (common customers in two queries) using INNER JOIN.

61. Write a query to simulate EXCEPT (customers in query A but not in query B) using NOT IN.

Topic 8: Query Execution Order

62. Explain why you cannot use an aggregate function directly inside a WHERE clause, referencing SQL execution order.

63. Write a query that uses WHERE, GROUP BY, HAVING, ORDER BY, and LIMIT together — combining smartphones data (e.g., 5G phones grouped by brand, having count > 5, sorted by avg price, limited to top 3).

Topic 9: Mixed / Practice (IPL-style dataset)

64. Find the top 5 batsmen by total runs.

65. Find the 2nd highest six-hitter.

66. Find a specific player's performance (total runs) against all opposing teams.

67. Find the top 10 batsmen with centuries (100+ runs in a single match).

68. Find the top 5 batsmen with the highest strike rate, among those who have faced a minimum of 1000 balls.